1. Run report: EXP-ECDLP-f49f75 (runs RUN-ECDLP-f49f75-01, −02, −03)

Observations only. This report records what ran and what was measured. It contains no verdict, support or weaken language: that belongs to the evidence review (`reports/review-DEC-20261015-a08389/`).

- **Experiment** EXP-ECDLP-f49f75 · **hypothesis** H-ECDLP-60de36 · **question** RQ-ECDLP-001 · **goal** GOAL-ECDLP-001
- **Launcher** `pkm_isogeny_fb.py`; **frozen plan** `trial-plan.json`; **amendments** AMD-01 (resume after host limit), AMD-02 (launcher int-overflow fix; invalidates RUN-01 m=2, L≥8 Gröbner cells, re-run as RUN-03).
- **Host** 4 vCPU, 15 GB, Linux 6.18, Python 3.13.16, Singular 4.3.2; 4 worker processes, one Singular process per solve.

1.1. Runs

run	cells	started (UTC)	finished	git HEAD	end
RUN-01	m=2, L=3..11 (L≥8 Gröbner excluded, AMD-02)	2026-10-15 04:59	06:59	378abed375	killed by 7200 s host limit in m=2 L=12 (no rows)
RUN-02	m=3, L=3..5	06:59	09:00	b9ddeefe78	killed by host limit in m=3 L=6 (no rows)
RUN-03	m=2, L=8..11	09:00	10:21	41a38de6c4	exit 0

Command (RUN-03): `python3 experiments/EXP-ECDLP-f49f75/pkm_isogeny_fb.py --plan experiments/EXP-ECDLP-f49f75/amendments/trial-plan-run03.json --out …/RUN-ECDLP-f49f75-03/raw-result.jsonl --workers 4`. Seeds: sha256 of `EXP-ECDLP-f49f75/v1|m|L|instance`. Per cell: 4 planted + 2 random targets. Timeout 600 s per Singular process. All working trees were clean at launch except the run directory being written.

1.2. Arms and controls

A: factor-base x-set $V = x(R + \langle K \rangle)$, $|V| = 2^L$, encoded by L chained Vélu 2-isogeny equations per point (mL variables). B: the same V by its degree-2^L minimal polynomial (m variables). C: a uniformly random 2^L-set, dense. E: meet-in-the-middle enumeration over ±FB (counted group operations). Solvers: Singular `std` and `slimgb` (each followed by `std` and `fglm` to lex). Positive control: the planted decomposition must lie in the computed variety.

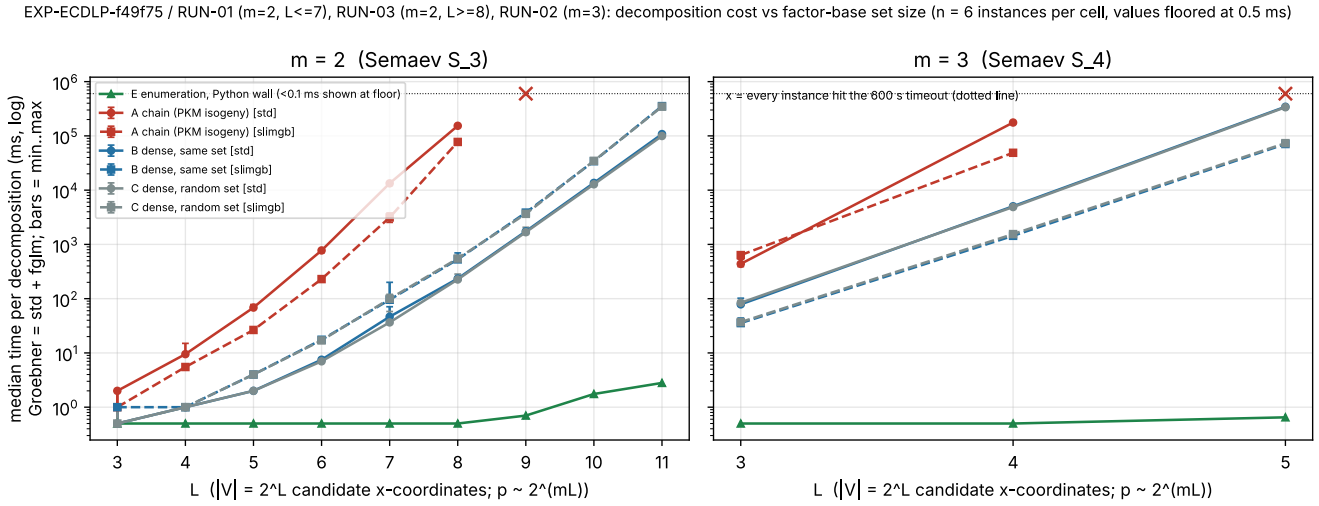


Figure 1: Median time per decomposition (Gröbner: std+fglm ms; E: Python wall, below 0.1 ms resolution shown at floor) against L; bars min–max over $n = 6$ instances (5 where a planted target was degenerate). × marks cells where every instance hit the 600 s timeout. Measured; RUN-01 (m=2, L≤7), RUN-03 (m=2, L≥8), RUN-02 (m=3).

1.3. Per-cell measurements

Columns: completed/timeout/degenerate-target counts; Gröbner median total ms [min–max] (measured) or enumeration median group ops and median factor-base size; planted-in-variety (passed/checked); max degree in reduced basis; max Singular memory(2) MiB.

m	L	p bits	run	arm	ok/to/deg	median ms [min–max] or ops	planted	maxdeg	MiB
2	3	7	R01	A	5/0/1	2 [1–2]	3/3	2	2
2	3	7	R01	A:slimgb	5/0/0	1 [1–2]	3/3	2	2
2	3	7	R01	B	5/0/1	0 [0–1]	3/3	4	2
2	3	7	R01	B:slimgb	5/0/0	1 [0–1]	3/3	4	2

2	3	7	R01	C	5/0/1	0 [0–1]	3/3	3	2
2	3	7	R01	C:slimgb	5/0/0	0 [0–1]	3/3	3	2
2	3	7	R01	E	5/0/1	8 ops / FB 4	—	—	—
2	4	9	R01	A	6/0/0	9.5 [9–15]	4/4	2	2
2	4	9	R01	A:slimgb	6/0/0	5.5 [5–6]	4/4	2	2
2	4	9	R01	B	6/0/0	1 [1–1]	4/4	3	2
2	4	9	R01	B:slimgb	6/0/0	1 [1–1]	4/4	3	2
2	4	9	R01	C	5/0/1	1 [1–1]	3/3	3	2
2	4	9	R01	C:slimgb	5/0/0	1 [1–1]	3/3	3	2
2	4	9	R01	E	6/0/0	17 ops / FB 8.5	—	—	—
2	5	11	R01	A	6/0/0	68.5 [65–78]	4/4	2	2
2	5	11	R01	A:slimgb	6/0/0	26.5 [26–30]	4/4	2	2
2	5	11	R01	B	6/0/0	2 [2–2]	4/4	3	2
2	5	11	R01	B:slimgb	6/0/0	4 [4–4]	4/4	3	2
2	5	11	R01	C	6/0/0	2 [2–2]	4/4	3	2
2	5	11	R01	C:slimgb	6/0/0	4 [4–4]	4/4	3	2
2	5	11	R01	E	6/0/0	27 ops / FB 13.5	—	—	—
2	6	13	R01	A	6/0/0	774 [706–802]	4/4	2	4
2	6	13	R01	A:slimgb	6/0/0	229.5 [223–240]	4/4	2	11
2	6	13	R01	B	6/0/0	7.5 [7–8]	4/4	3	2
2	6	13	R01	B:slimgb	6/0/0	17.5 [17–19]	4/4	3	2
2	6	13	R01	C	6/0/0	7 [7–8]	4/4	2	2
2	6	13	R01	C:slimgb	6/0/0	17 [17–18]	4/4	2	2
2	6	13	R01	E	6/0/0	65 ops / FB 32.5	—	—	—
2	7	15	R01	A	6/0/0	13,344 [12,193–13,819]	4/4	2	17
2	7	15	R01	A:slimgb	6/0/0	3,138.5 [2,547–3,796]	4/4	2	58
2	7	15	R01	B	6/0/0	46.5 [35–71]	4/4	4	4
2	7	15	R01	B:slimgb	6/0/0	95.5 [92–200]	4/4	4	6
2	7	15	R01	C	6/0/0	36.5 [34–58]	4/4	5	4
2	7	15	R01	C:slimgb	6/0/0	99.5 [92–120]	4/4	5	6
2	7	15	R01	E	6/0/0	124 ops / FB 62	—	—	—
2	8	17	R03	A	6/0/0	153,435 [144,479–158,329]	4/4	2	59
2	8	17	R03	A:slimgb	6/0/0	77,357 [71,380–78,300]	4/4	2	90
2	8	17	R03	B	6/0/0	234.5 [228–250]	4/4	3	14
2	8	17	R03	B:slimgb	6/0/0	529.5 [480–697]	4/4	3	16
2	8	17	R03	C	6/0/0	225.5 [215–284]	4/4	2	14
2	8	17	R03	C:slimgb	6/0/0	549.5 [496–568]	4/4	2	16
2	8	17	R03	E	6/0/0	273 ops / FB 136.5	—	—	—
2	9	19	R03	A	0/6/0	—	—	—	—
2	9	19	R03	A:slimgb	0/6/0	—	—	—	—
2	9	19	R03	B	6/0/0	1,743 [1,596–1,820]	4/4	3	50
2	9	19	R03	B:slimgb	6/0/0	3,855 [3,646–4,341]	4/4	3	64
2	9	19	R03	C	6/0/0	1,678 [1,530–2,067]	4/4	3	50
2	9	19	R03	C:slimgb	6/0/0	3,658.5 [3,494–3,894]	4/4	3	64
2	9	19	R03	E	6/0/0	514 ops / FB 257	—	—	—
2	10	21	R03	B	6/0/0	13,604.5 [12,681–14,261]	4/4	3	196
2	10	21	R03	B:slimgb	6/0/0	34,144.5 [32,964–34,620]	4/4	3	255
2	10	21	R03	C	6/0/0	12,801.5 [12,660–13,429]	4/4	4	196
2	10	21	R03	C:slimgb	6/0/0	34,614.5 [33,650–35,094]	4/4	4	255

2	10	21	R03	E	6/0/0	1,045 ops / FB 522.5	—	—	—
2	11	23	R03	B	6/0/0	107,962.5 [101,522–110,205]	4/4	3	783
2	11	23	R03	B:slimgb	6/0/0	353,720.5 [345,233–355,022]	4/4	3	1013
2	11	23	R03	C	6/0/0	99,491 [98,921–99,982]	4/4	3	783
2	11	23	R03	C:slimgb	6/0/0	345,917.5 [338,619–351,202]	4/4	3	1013
2	11	23	R03	E	6/0/0	2,028 ops / FB 1,014	—	—	—
3	3	10	R02	A	6/0/0	437.5 [390–535]	4/4	3	8
3	3	10	R02	A:slimgb	6/0/0	635.5 [601–729]	4/4	3	29
3	3	10	R02	B	6/0/0	78.5 [75–102]	4/4	8	4
3	3	10	R02	B:slimgb	6/0/0	35.5 [32–44]	4/4	8	4
3	3	10	R02	C	6/0/0	84 [78–92]	4/4	3	4
3	3	10	R02	C:slimgb	6/0/0	37 [31–40]	4/4	3	4
3	3	10	R02	E	6/0/0	54.5 ops / FB 4.5	—	—	—
3	4	13	R02	A	6/0/0	176,155 [168,125–179,709]	4/4	2	235
3	4	13	R02	A:slimgb	6/0/0	48,804 [45,650–50,569]	4/4	2	746
3	4	13	R02	B	6/0/0	5,049 [4,703–5,363]	4/4	3	40
3	4	13	R02	B:slimgb	6/0/0	1,439.5 [1,337–1,515]	4/4	3	39
3	4	13	R02	C	6/0/0	4,882.5 [4,783–5,088]	4/4	16	40
3	4	13	R02	C:slimgb	6/0/0	1,548.5 [1,332–1,631]	4/4	16	39
3	4	13	R02	E	6/0/0	152 ops / FB 8	—	—	—
3	5	16	R02	A	0/6/0	—	—	—	—
3	5	16	R02	A:slimgb	0/6/0	—	—	—	—
3	5	16	R02	B	6/0/0	344,179.5 [333,653–348,474]	4/4	3	615
3	5	16	R02	B:slimgb	6/0/0	70,800.5 [67,665–72,581]	4/4	3	459
3	5	16	R02	C	6/0/0	337,273.5 [334,025–339,216]	4/4	4	615
3	5	16	R02	C:slimgb	6/0/0	73,460.5 [65,149–76,113]	4/4	4	459
3	5	16	R02	E	6/0/0	629 ops / FB 17	—	—	—

1.4. Operational notes (facts, not findings)

- 24 RUN-01 rows (m=2, L=8..11) are excluded under AMD-02: their Singular input evaluated $x(T)^2$ in 32-bit integer arithmetic for $p > 46340$; the planted control fell to 2/4, 1/4, 0/4, 0/4 there. RUN-03 re-ran these cells with the fixed launcher; every planted check passed.
- RUN-02 workers were forked before the AMD-02 launcher fix and therefore ran launcher commit b9ddeefe78; the m=3 input never squares a bare integer and all its planted checks passed.
- Four AMD-02 fix-verification solves (07:15–07:20 UTC) shared the host with RUN-02′s m=3, L=3 cell; its timings may be inflated.
- Degenerate targets (planted sum = point at infinity): 1 per affected cell, recorded as such.
- Arm A retired by the frozen stopping rule at m=2, L=9 and at m=3, L=5. m=2, L=12 and m=3, L=6 were not executed (host limit; AMD-01).